[SECURITY AUDIT] MCCJ 525 · CYBERCRIMES AND SECURITY

Locked or Leaked?

Assessing the Security Features of Facebook, Instagram, WhatsApp, Gmail and Telegram

> scanning 5 platforms ··· 5 criteria ··· 25 points ··· DONE



ENCRYPTED

PRESENTED BY

[Member 1] · [Member 2] · [Member 3] · [Member 4] · [Member 5]

Group [__] | [Department], [University] | [Date]

Account hacking is the most common cybercrime in Bangladesh

21.65%

of victims had a social media or online account hacked, the most common offence

≈59%

of cybercrime victims are women

78.78%

of victims are aged 18–30, the age group in this room

12%

of victims reported the incident to law enforcement

≈ 64 million Bangladeshis use Facebook. Attackers no longer break into computers; they walk into accounts.

Sources: Cyber Crime Awareness Foundation (2024), Cybercrime Trends in Bangladesh, reported by Dhaka Tribune; Statista (2025).

We assessed five platforms against five security criteria

01 EXPLORE
Reviewed every security setting on our own accounts

02 VERIFY
Cross-checked with official help pages and recent news

03 SCORE
Rated each criterion 1 (weak) to 5 (strong); maximum 25

C01

Login protection

Can a stolen password alone get someone in?

C02

Encryption by default

Can the company itself read my messages?

C03

Threat alerts

Will the platform warn me of suspicious activity?

C04

Recovery safety

Can a hacker lock me out permanently?

C05

Privacy controls

Can strangers find, view or contact me?

Facebook: strong login tools, but weak account recovery

EXISTING SECURITY FEATURES

Two-factor authentication
Authenticator app, SMS code or physical security key

Login alerts and remote log-out
“Where you’re logged in” in Accounts Center

Passkeys (June 2025)
Fingerprint or face scan replaces the password

Profile lock
Hides photos and posts from non-friends

Encrypted Messenger
Personal chats end-to-end encrypted by default since Dec 2023

SECURITY SCORE

16 /25

JOINT 3RD



KEY GAP

Hackers can change the recovery email and phone number within minutes, and recovery is slow and largely automated.

RECOMMENDED FIX

A 48-hour cool-down on recovery changes, with alerts to old contacts, plus Bangla-language human support.

Sources: Meta (2025); Crisan (2023).

Instagram: useful anti-phishing tools, but DMs are not encrypted

EXISTING SECURITY FEATURES

Meta Accounts Center security
Same 2FA, login alerts and login activity as Facebook

“Emails from Instagram”
Lists genuine security emails from the last 14 days

Restrict and Hidden Words
Filters abusive comments and message requests

Teen Accounts
Private by default for users under 18

SECURITY SCORE

16 /25

JOINT 3RD



KEY GAP

DMs were never encrypted by default, and Meta removed the optional encrypted chats on 8 May 2026.

RECOMMENDED FIX

Restore end-to-end encrypted DMs as the default, and blur unsolicited explicit images for all users.

Source: Euronews (2026).

WhatsApp: encrypted by default, but exposed to code-sharing scams

EXISTING SECURITY FEATURES

- **End-to-end encryption by default**
  All personal chats, calls and status updates
- **Two-step verification PIN**
  Six-digit PIN required to re-register the number
- **Passkeys and encrypted backups**
  Backups can be locked with a passkey (Oct 2025)
- **Privacy controls**
  Chat lock, silence unknown callers, IP protection in calls
- **Strict Account Settings (Jan 2026)**
  One-tap lockdown against unknown senders

SECURITY SCORE

20 /25

JOINT 1ST



KEY GAP

The phone number is the identity; users tricked into sharing the six-digit registration code lose the account.

RECOMMENDED FIX

A Bangla warning whenever a registration code is requested, and a mandatory two-step PIN.

Sources: WhatsApp (2026); Help Net Security (2025).

Gmail: the best account protection, guarding the master key

EXISTING SECURITY FEATURES

- **Passkeys by default (Oct 2023)**
  Password-free sign-in for personal Google accounts
- **2-Step Verification and Security Checkup**
  Guided review of sign-in methods and devices
- **Critical security alerts**
  Warnings of suspicious sign-ins and recent activity
- **Spam and phishing filters**
  Google reports over 99.9% of phishing and malware blocked
- **Advanced Protection Program**
  Hardware-key protection for high-risk users

SECURITY SCORE

20 /25

JOINT 1ST



KEY GAP

Personal Gmail is not end-to-end encrypted, and it controls password resets for every other account.

RECOMMENDED FIX

Optional end-to-end encryption for personal mail, and lookalike warnings for bKash, Nagad and local banks.

Sources: Google (2023); Engadget (2025).

Telegram: its privacy reputation exceeds its default settings

EXISTING SECURITY FEATURES

Two-step verification
Additional cloud password for new logins

Passcode lock
Local lock on the app itself

Active sessions
View and terminate other logged-in devices

Phone number privacy
Hide the number entirely; connect by username

Secret Chats
End-to-end encrypted, device-specific, self-destruct timers

SECURITY SCORE

14 /25

5TH



KEY GAP

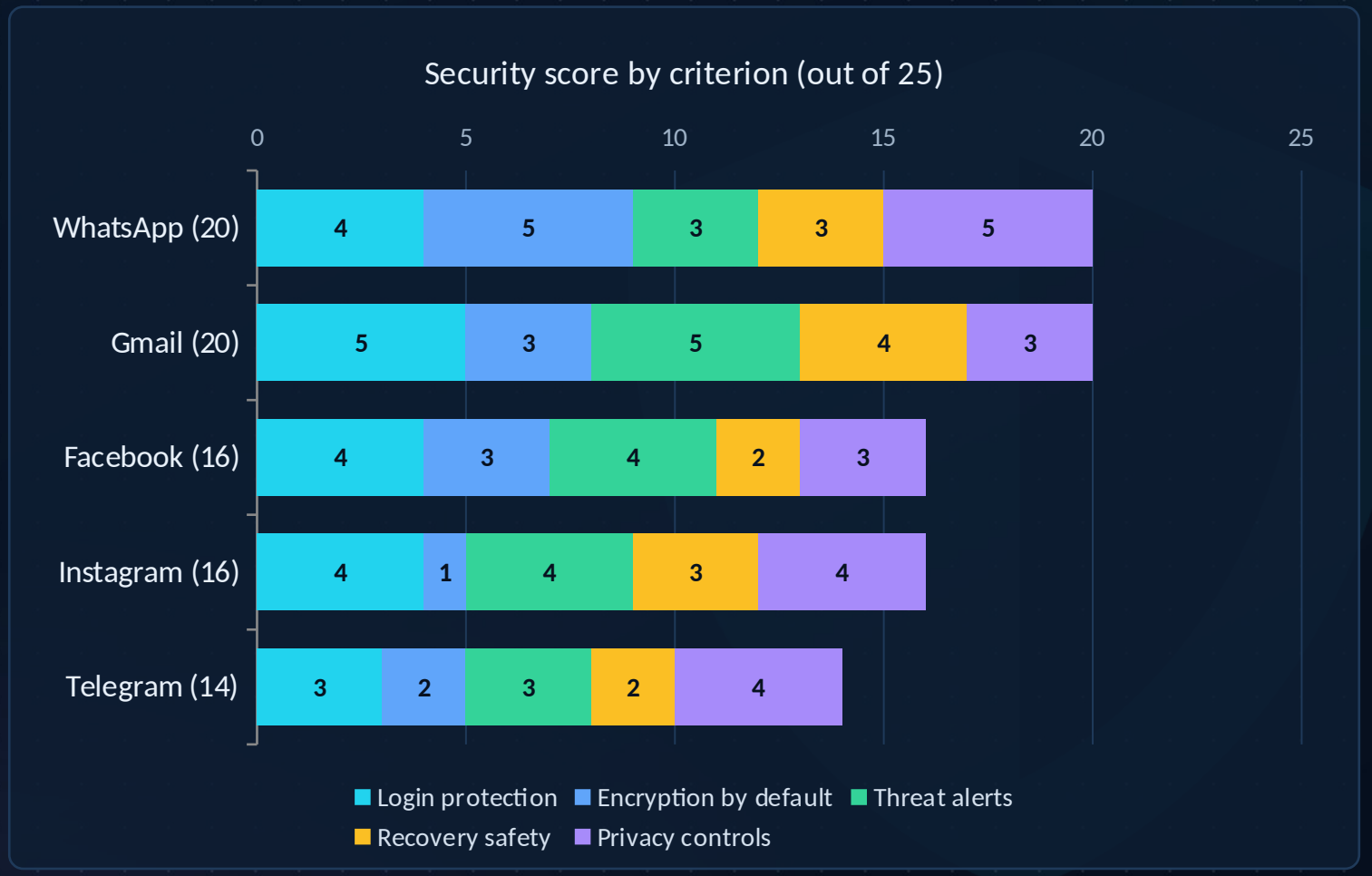
Regular chats, groups and channels are not end-to-end encrypted, and since Sept 2024 IPs and phone numbers may be shared on valid legal request.

RECOMMENDED FIX

End-to-end encryption by default for one-to-one chats, and a two-step verification prompt at sign-up.

Source: Engadget (2024).

WhatsApp and Gmail lead; WhatsApp is the most secure by default



Most secure for individuals: WhatsApp

Its strongest protection, end-to-end encryption, works from installation with no setup.

Highest-value account: Gmail

Best login protection and filtering; it resets every other password.

Lowest score: Telegram

Ordinary chats are not end-to-end encrypted by default.

Scores reflect the group's assessment of default and available settings, October 2026.

// 09 ACTION PLAN

The most secure app is the one you set up

> YOUR FIVE-MINUTE SECURITY CHECKLIST

[01]

Enable two-factor authentication or a passkey on every account

[02]

Set a two-step verification PIN on WhatsApp

[03]

Never share an OTP or six-digit code with anyone

[04]

Confirm your recovery email and phone number

[05]

Log out of any device you do not recognise

Thank you | Questions and discussion

SESSION CLOSED